COL724/COL7524: Advanced Computer Networks

Semester 1, 2026–27

Project Guidelines

1. Finding a Research Topic

Find a relevant paper published that satisfies the following criteria:

1. It should have been published after the year 2020.
2. It should not be a survey paper or a systematization-of-knowledge (SoK) paper.

It is advised to use Google Scholar when searching for research papers and similar scholarly material.

2. Selecting a Research Paper

There could be a variety of ways to select the topic and the relevant paper for your project. Please feel free to find your way. The following could be potential starting points:

- Identify emerging trends or issues in the field of networks.
- Look for open problems or challenges discussed in recent publications.
- Find the program of any of the top conferences on their websites and read through the session titles.
- If you find a session title interesting, read the paper titles.
- If you find a paper title interesting, read its abstract.
- If you find the abstract interesting, read the captions of the figures and tables.
- If you still find the paper interesting, read the introduction.
- Some of the top conferences are ACM SIGCOMM, ACM MobiCom, USENIX NSDI, IEEE INFOCOM, ACM IMC, NDSS, ACM SOSP.

It is advised to read a research paper in multiple iterations so as to get the most sense out of it. For example:

- In the first iteration, simply skim through the paper.
- Mark topics or figures that you find interesting or that you need to learn more about.
- Complete your prerequisites, and then make a more thorough read of the paper.

Once you have read some papers, you should be able to judge whether you would be interested in carrying out the project work in that area.

Note: In addition to selecting the research paper, students are required to check the subsequent work related to the selected paper, including papers published later by the same authors or other researchers that build upon, extend, or improve the original work. Students should identify what additional work, modifications, or extensions have already been proposed in these subsequent papers. The proposed project work must be substantially different from the extensions or contributions already presented in the related subsequent work. Students should clearly demonstrate that their proposed contribution is novel and does not simply reproduce or duplicate work that has already been carried out.

3. Finalizing the Project

Once you have decided your project and paper, please fill in your entry in the shared sheet by **20th August 2026**. Topic selection will be prioritized on a First-come-first-serve (FCFS) basis.

- You are allowed to do the project in groups of size at most 2.
- No two groups should work on the same paper/project/topic.
- You will not be allowed to change your groups after 20th August 2026 as the entries will be locked.
- Copying previous year projects is strictly prohibited and will be treated as a case of plagiarism.

4. Submissions on Moodle

You are expected to make the following two PDF submissions over Moodle:

1. Paper Abstract

A brief abstract of the paper selected by you and what contributions/extensions you think can be made to that paper. Please write the abstract in your own words. Do not copy the same abstract from the paper.

Deadline: 20th August 2026

2. Paper Report

You are expected to propose the work that you plan to carry out as part of the project. Please submit a PDF file containing the following:

1. Group Details (Names and Entry Numbers)

2. Paper Details (Publication Date, Conference Details, Number of Citations, etc.)
3. Key Takeaways
4. Project Title
5. Problem Statement
6. Proposed Solutions/Extensions

Deadline: 20th August 2026

Submission Format: Submit a single .zip file with the following naming convention:

`<entryNumber1>_<entryNumber2>_projectProposal.zip`

If you are working individually, use only your entry number, for example:

`csz1234_projectProposal.zip`

5. Project Presentations

The project will include **three presentations** at different stages of the semester. Each presentation should demonstrate the progress of the project and should be prepared according to the corresponding stage described below.

1. Initial Paper Selection and Proposal Presentation

The first presentation will focus on the selected research paper and the proposed direction for the project.

The presentation should include:

1. Introduction to the selected research paper.
2. Brief presentation and explanation of the paper, including its motivation, problem statement, methodology, and key results.
3. Abstract of the paper written in your own words.
4. Conference details, including the conference in which the paper was published.
5. The selected paper should be from a **top conference** in the relevant area.
6. Key observations, limitations, or open problems identified in the paper.
7. Your proposed contribution, modification, extension, or additional work that you intend to carry out based on the paper.

The paper selection, abstract, and proposed project direction must be finalized and submitted by **20th August 2026**. The first presentation will be held on **24th August 2026**. Students should prepare the presentation and associated material accordingly.

2. Progress Presentation

The second presentation will be conducted **after the Minor Examination**. The purpose of this presentation is to evaluate the progress made on the project after the initial paper selection and proposal.

The presentation should include:

1. A brief recap of the selected paper and the project objective.
2. Work completed since the first presentation.
3. Implementation, experiments, simulations, or theoretical work carried out so far, as applicable.
4. Preliminary results and observations.
5. Challenges or difficulties encountered during the project.
6. Any changes or refinements made to the original project proposal.
7. Planned work and milestones for completing the remaining project.

3. Final Presentation and Report

The third presentation will be conducted **before the Major Examination**. This will be the final presentation of the complete project.

The presentation and report should include:

1. Complete introduction and motivation of the project.
2. Detailed explanation of the selected research paper.
3. Problem statement and research objectives.
4. Detailed description of the proposed contribution, extension, or modification.
5. Methodology and implementation details.
6. Experimental setup, evaluation, or theoretical analysis, as applicable.
7. Results and detailed discussion.
8. Comparison with the original paper or existing approaches, wherever applicable.
9. Limitations and possible future work.
10. Complete project conclusions.

A **complete project report** must also be submitted along with the final presentation. The final presentation and report should reflect the overall work carried out during the semester and incorporate the feedback received during the earlier presentations.